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Produced Water Reuse for Sustainable SmartWater Flooding in Carbonates

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Abstract

SmartWater flooding through tailoring of injection water salinity and ionic composition has gained significant attention for improved oil recovery in carbonates. In this paper, we experimentally evaluate the potential of reusing produced water to achieve sustainable SmartWater flooding in carbonates. For this purpose, we utilized the treated low salinity waters obtained from the desalination of hypersaline produced water and unconventional frac flow-back produced water using zero liquid discharge (ZLD) technology to generate desired SmartWater chemistries.

Three types of tailored SmartWater recipes were formulated using high salinity water and two treated low salinity produced waters. First SmartWater (conventional) represents the diluted high salinity water version with deionized water, while the other two SmartWater formulations are obtained by blending of high salinity water with treated low salinity produced water streams. Both zeta potential and core flooding experiments were conducted using carbonate rocks and crude oil. Zeta potentials were measured in calcite / brine, carbonate rock /brine, and crude oil in brine suspensions at both ambient and elevated temperatures (25 °C and 70 °C). Core flooding experiments were carried out at reservoir conditions (70 °C and 2000 psi) by first injecting high salinity water then followed by the injection of different SmartWater chemistries in tertiary mode.

The calcite/brine and carbonate rock/brine zeta potentials showed negative surface charges with all SmartWater chemistries. Interestingly SmartWater formulations obtained with treated produced water showed increasingly negative surface charges at calcite/brine and rock/brine interfaces when compared to conventional SmartWater. The measured zeta potentials at crude oil/brine interfaces also showed slightly higher negative surface charges for both the treated produced water based SmartWater chemistries. The same zeta potential trends were observed with SmartWaters at elevated temperature across calcite/brine, rock/brine and oil/brine interfaces even though the magnitude of negative surface charges were found to be relatively higher over those obtained at ambient temperature. These zeta potential results demonstrate that SmartWaters prepared with treated produced water have enhanced wettability alteration capability over conventional SmartWater formulated using high salinity water. The core flood performed with conventional SmartWater showed about 7% incremental oil recovery beyond high salinity water injection, while the core flood incremental oil recoveries were about 16% and 13% with the injection of SmartWaters formulated using the two treated produced waters, respectively. The core flood results corroborated very well with

zeta potential data to confirm the potential of reusing treated produced water to formulate SmartWater and increase oil recovery in carbonates.

This experimental study for the first-time demonstrates the potential of reusing treated produced water as effective "wettability modifier" to achieve sustainable SmartWater flooding in carbonates. The recycling of produced water as 'SmartWater' can become a game changer to avoid costly produced water disposal and promote circular water economy/environmental sustainability in SmartWater flooding field projects.

Introduction

Waterflooding is a widely used secondary recovery technique to improve oil recovery, particularly effective in carbonate reservoirs, which hold over 50% of global oil reserves. These formations are geologically complex, with high heterogeneity in porosity and permeability, making oil displacement challenging. Traditional waterflooding often leaves considerable oil behind, primarily due to oil-wet surface conditions. This limitation has driven interest in more advanced water flooding methods like SmartWater flooding (Yousef et al., 2011; Rashid et al., 2015; Sohal et al., 2016; Ayirala et al. 2018; Abu-Al-Saud et al., 2020).

SmartWater flooding involves optimizing the salinity and ionic makeup of injection water to influence the electrokinetic interactions at the rock–fluids interface. By modifying zeta potential, SmartWater alters the wettability of the rock surface, promoting a more water-wet environment and improving oil recovery (Alghamdi et al., 2020; Ayirala and Yousef, 2016; Yousef and Ayirala, 2014). This process is especially beneficial in carbonates, where oil-wet surfaces typically reduce recovery efficiency.

Numerous experimental studies have evaluated SmartWater's potential for improved oil recovery in carbonates. Coreflooding experiments, which simulate reservoir conditions, are instrumental in assessing oil recovery and flow dynamics. These studies have consistently demonstrated that tailored brine compositions can improve oil displacement by altering the rock's wettability (Yousef et al., 2012; Ayirala et al., 2016). Zeta potential measurements serve as a key indicator of surface charge behavior and are closely linked to fluid-rock interactions. Adjustments to ionic concentration—particularly the balance between monovalent and divalent ions have shown measurable impacts on zeta potential, correlating with improved oil recovery (Jackson et al., 2016).

Incorporating SmartWater into conventional flooding practices introduces a more efficient, water chemistry-driven enhanced oil recovery (EOR) method. By carefully tuning the ionic composition of the injected brine, it becomes possible to reduce residual oil saturation and enhance recovery in carbonate reservoirs. If we can utilize SmartWaters obtained from produced water to increase oil recovery, it becomes a sustainable approach to recycle produced water and promote circular water economy in upstream operations.

In this study, we investigate the effect of different SmartWater chemistries formulated using conventional high salinity water and treated produced waters on oil recovery and surface charge characteristics in carbonate-brine-oil systems. The two different treated produced waters used in this study were obtained from produced water desalination using zero liquid discharge (ZLD) technology (Ayirala et al., 2024; Muhammad et al., 2024) comprising of a customized pretreatment system to lower dissolved H_2S /oil and dynamic vapor recovery unit to remove salts from produced water. Zeta potential and coreflooding experiments were conducted to assess wettability alteration, fluid-rock interactions, incremental oil recovery, and eventually demonstrate the viability of using recycled produced water as part of developing sustainable SmartWater flooding strategies for the industry.

Experimental Materials, Methods, and Procedure

This section presents a comprehensive description of the experimental approach undertaken to evaluate the coreflooding oil recoveries and interfacial electrokinetic properties of various SmartWater formulations. It begins with an overview of the rock samples, brine chemistries, and fluid characteristics employed in the study. The subsequent subsections detail the experimental apparatus configuration and procedural steps

used for coreflooding tests, followed by the methodology adopted for zeta potential measurements. Both the corefloods and electrokinetic measurements were conducted using three distinct brine formulations: Conventional SmartWater 1 (10-times diluted high salinity water), SmartWater 2 based on Treated Produced Water 1 (TPW1), and SmartWater 3 obtained using Treated Produced Water 2 (TPW2).

Rock Samples

High-purity calcite disks containing approximately 99 wt.% CaCO_3 were used for zeta potential measurements and coreflooding experiments were performed using carbonate cores. These mineralogical compositions were determined using X-ray diffraction (XRD) as shown in the Figure 1. The calcite samples were manually crushed and ground using a granite mill to achieve fine particle sizes suitable for zeta potential experiments. The core samples of approximately 2 inch (length) and 1.5 inch (diameter) dimensions were used for core flooding experiments. The properties of core samples including the dimensions, porosity, permeability, and pore volumes are summarized in Table 1.

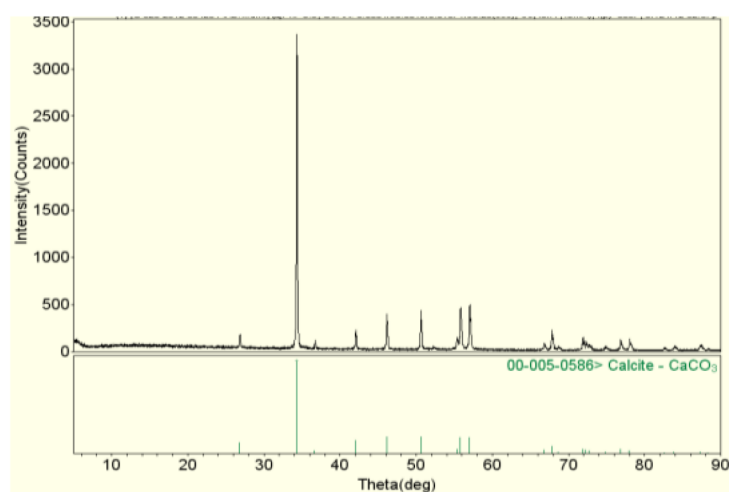


Figure 1—X-ray diffraction (XRD) pattern of high-purity calcite (CaCO_3) sample used in zeta potential experiments.

Table 1—Basic petrophysical properties of the core samples used for core flooding experiments.

Sample	Diameter (cm)	Length (cm)	Porosity (fraction)	Air Permeability (mD)	Pore Volume (cm^3)
SmartWater 1	3.74	4.55	0.28	513.2	14.0
SmartWater 2 (TPW1)	3.76	4.44	0.27	542.9	14.5
SmartWater 3 (TPW2)	3.76	4.44	0.28	524.5	14.7

Crude Oil Properties

The crude oil employed in this study was obtained from a carbonate reservoir. Prior to use, it was purified to remove suspended solids and contaminants through centrifugation and standard extraction techniques. The properties of crude oil sample are summarized in Table 2.

Table 2—Properties of the crude oil used in this study after purification and preparation.

Property	Value
API Gravity	28.3
Viscosity @ Ambient Conditions	6.0 cP
Density @ 25 °C	0.8845 g/cm ³
Acid Number AN	0.26 mg KOH/g

Brine Compositions

The synthetic brines used in this study were prepared using high-purity analytical-grade salts. The compositions of the brine samples, including High Salinity Water (HSW), SmartWater, TPW1, and TPW2, are summarized in Table 3. SmartWater 1 is prepared by diluting high salinity water 10-times using deionized water. The two treated produced waters, TPW 1 and TPW 2, were obtained from produced water desalination using zero liquid discharge (ZLD) technology by processing hypersaline produced water (80,000 ppm TDS) and unconventional frac flow back produced water (greater than 250,000 ppm TDS), respectively. Smarter 2 is prepared by diluting high salinity water with TPW 1 by 10-times, while TPW 2 has been used for 10-times dilution of high salinity water to obtain SmartWater 3. These specific SmartWater formulations were chosen to investigate their influence on surface charge behavior, wettability alteration, and improved oil recovery in carbonate rocks.

Table 3—Brine compositions used in this study.

<i>Brine Compositions (concentration mg/L)</i>				
Component	High Salinity Water (HSW)	SmartWater 1	TPW 1	TPW 2
Na ⁺	18,900	1,824	<1.0	<1.0
Cl ⁻	32,200	3,220	48	112
Ca ²⁺	650	65	<1.0	<1.0
Mg ²⁺	210	21	<1.0	<1.0
SO ₄ ²⁻	4,920	492	122	129
HCO ₃ ⁻	120	12	<1.0	<1.0
Total Dissolved Solids (TDS)	57,670	5,761	374	402
pH	7.4	7.4	9.1	9.8

Coreflooding Experiments

Carbonate core samples were prepared by saturating them with crude oil under vacuum for 2 days to ensure full pore saturation and eliminate air entrapment. The cores were then aged at 70 °C for 3 weeks to restore wettability and develop oil-wet characteristics typical of carbonate reservoirs.

The coreflooding experiments were performed using a high-pressure core flooding apparatus equipped with a stainless-steel core holder. The schematic of core flood apparatus is shown in Figure 2. A confining pressure of 2000 psi was applied to simulate overburden stress. Crude oil was injected at a constant flow rate of 0.25 cc/min under a controlled pore pressure of 1200 psi. Injection continued until 8 pore volumes (PV) had passed through each core to ensure steady-state oil saturation. Back pressure was adjusted as needed to maintain stable flow conditions throughout the experiment.

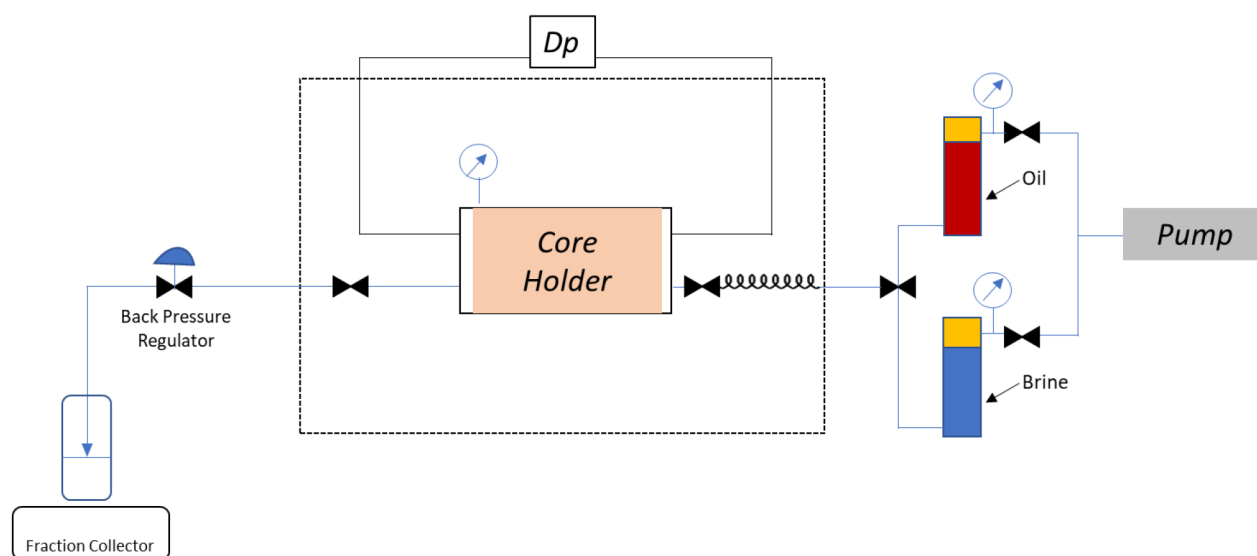


Figure 2—Schematic diagram of the coreflooding setup used in this study.

The injection sequence for all oil displacement tests included two main stages:

- Stage 1 – Baseline Water Flooding: Each core was initially injected with 5 PV of conventional high salinity water (HSW) to simulate secondary water flooding and establish a baseline residual oil saturation for tertiary stage oil displacement by different SmartWater formulations.
- Stage 2 – SmartWater Injection:
 - Conventional SmartWater/SmartWater 1: 5 PV of diluted high-salinity brine.
 - SmartWater 2: 5 PV of a blend containing 90% treated produced water 1 and 10% high salinity water
 - SmartWater 3: 5 PV of a blend containing 90% treated produced water 2 and 10% high salinity water.

During each stage, oil recovery (as % of original oil in place, OOIP) and pressure differential across the core were continuously monitored to evaluate the performance of each injection SmartWater to mobilize residual oil and result in enhanced oil recovery.

Zeta Potential Measurements

Zeta potential measurements were performed to assess the surface charge characteristics and wettability alteration potential at three interfaces: calcite/brine, carbonate rock/brine, and crude oil/brine. These tests were conducted at two temperatures, 25 °C and 70 °C, to simulate both ambient and elevated temperature conditions.

The measurements were carried out using an electrophoretic light scattering instrument, which determines the zeta potential based on the electrophoretic mobility of particles suspended in brine when subjected to an external electric field. The Smoluchowski equation was applied to convert the measured mobility into zeta potential values. This approach is widely adopted to measure zeta potentials in rock/brine and crude oil/brine systems (Alghamdi et al. 2021). The photograph of experimental setup used for zeta potential measurements is shown in Figure 3.



Figure 3—Zeta PALS analyzer used for measuring the zeta potential of particle suspensions.

For the calcite/brine and carbonate rock/brine measurements, powdered samples (0.2 g) were dispersed in 50 mL of each test brine. The suspensions were sonicated and then aged for 48 hours to ensure equilibrium between the solid surface and brine chemistry. In the crude oil/brine case, emulsions were prepared by mixing crude oil and brine to form stable droplets in the range of 14 to 35 micron sizes. After equilibrium aging, samples were centrifuged to eliminate unstable or aggregated particles before measurement.

To ensure the accuracy and reliability of the results, each zeta potential measurement was repeated five times, and the mean value was reported. The standard deviation of the repeated readings was maintained within ± 3 mV, indicating high reproducibility. Temperature control during measurement was achieved using a thermostatic cell holder to ensure stable temperatures during the experiments.

Results and Discussion

SmartWater effects on zeta-potentials

Zeta potential measurements were conducted to evaluate the impact of different SmartWater formulations on the surface charge behavior of calcite/brine, carbonate/brine and crude oil/brine systems. The tests were performed at both ambient (25 °C) and elevated (70 °C) temperatures across three interfaces: calcite/brine, carbonate rock/brine, and crude oil/brine. These measurements help clarify the role of SmartWater on surface charge alteration in promoting wettability change, which is a key mechanism in enhancing oil recovery. The zeta potential results obtained with different SmartWater formulations across calcite/brine, carbonate rock/brine, and crude oil/brine interfaces at 25 and 70°C are shown in Figure 4 and 5, respectively.

At 25 °C, conventional SmartWater (SmartWater 1), formulated by diluting high salinity brine, exhibited moderately negative zeta potentials. The values recorded were -5.58 mV for the calcite/brine interface, -9.56 mV for the carbonate rock/brine interface, and -0.17 mV for the crude oil/brine interface. When compared to conventional SmartWater, the treated produced water-based Smart Waters (Smart Water 2 and Smart Water 3) demonstrated increasingly more negative surface charges, particularly at the rock/brine interfaces. Smart Water 2 showed zeta potentials of -14.37 mV (calcite/brine), -15.09 mV (carbonate rock/brine), and -0.44 mV (crude oil/brine), while Smart Water 3 yielded even more negative values: -19.63 mV, -21.16 mV, and -0.68 mV, respectively, at different interfaces.

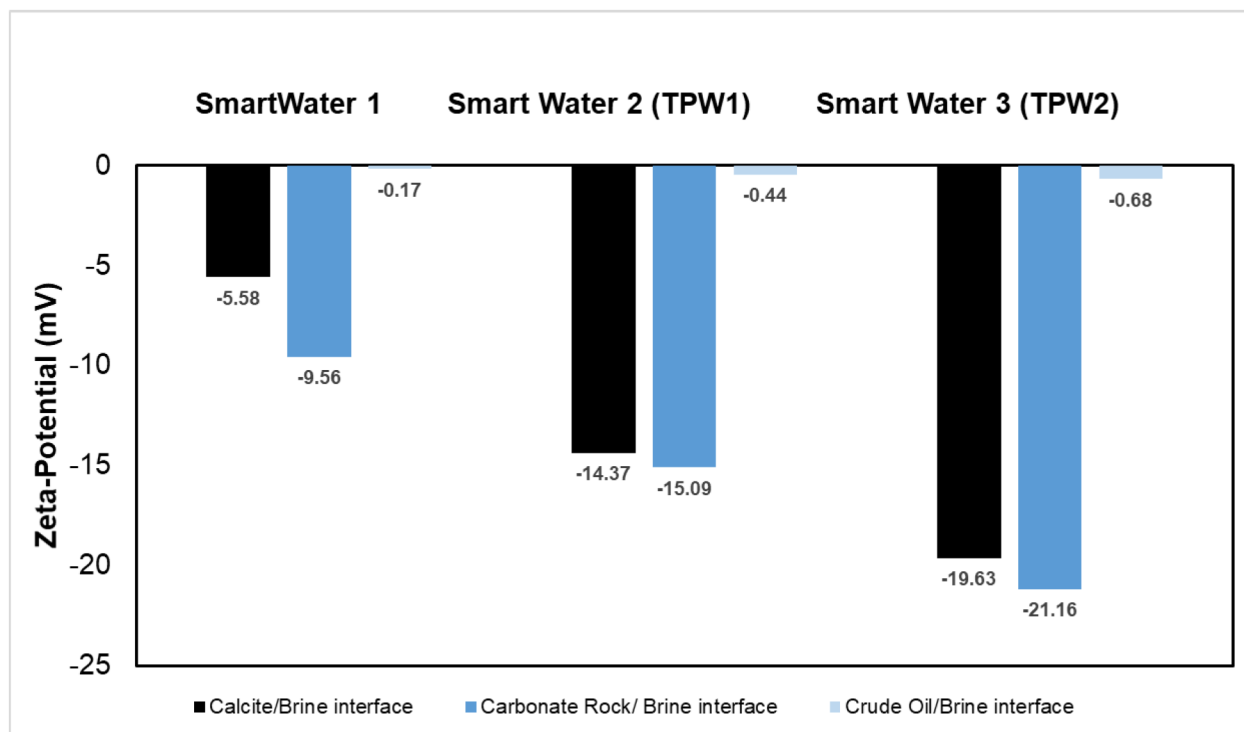


Figure 4—Zeta potential measurements at 25 °C for different brine formulations at the calcite/brine, carbonate rock/brine, and crude oil/brine interfaces.

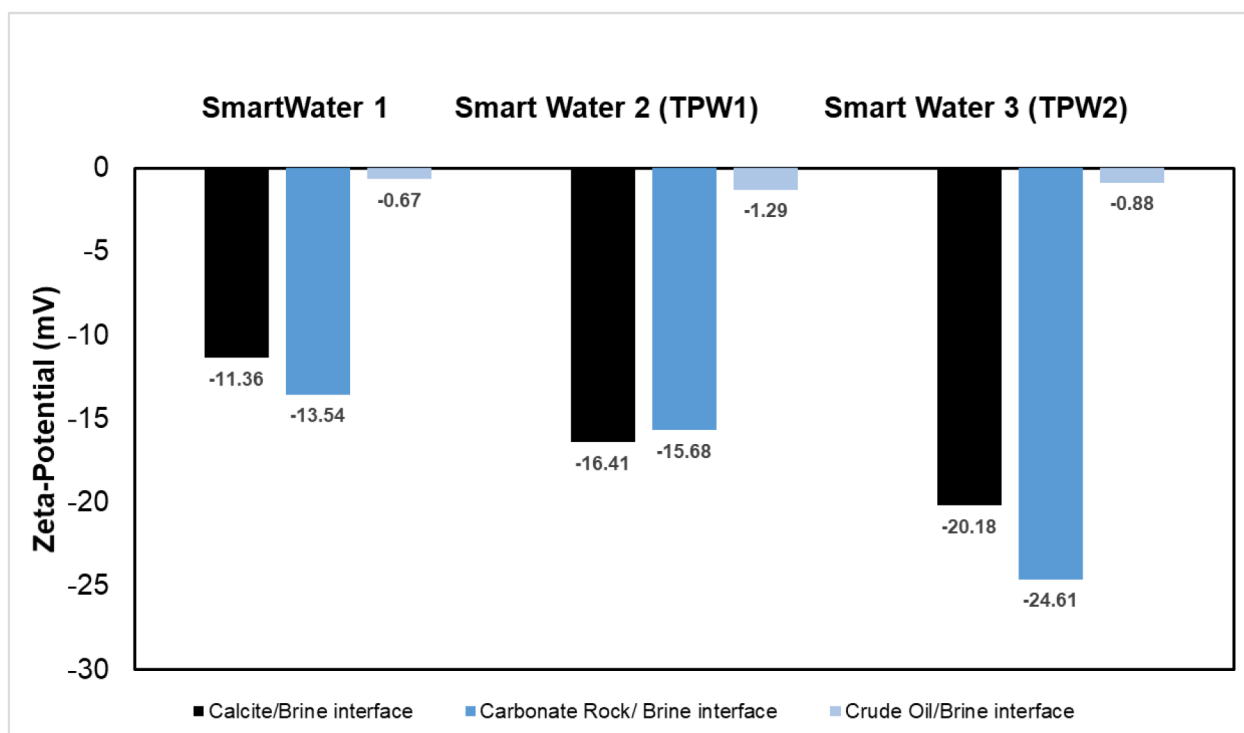


Figure 5—Zeta potential measurements at 70 °C for different brine formulations at the calcite/brine, carbonate rock/brine, and crude oil/brine interfaces.

The influence of temperature was also found to be substantial. At 70 °C, three SmartWater brines showed an increase in the magnitude of negative surface charges across all interfaces. For instance, the zeta potential of Smart Water 2 reached -20.18 mV at the calcite/brine interface and -24.61 mV at the carbonate rock/

brine interface. These results confirm that temperature increase enhances the electrostatic repulsion at rock/brine and crude oil-brine interfaces, likely by increasing ion mobility and expanding the electrical double layer (Nasralla and Nasr-EI-Din, 2014). Even the crude oil/brine interface showed more negative values at higher temperatures, indicating much stronger electrostatic repulsion between the two interfaces and thereby signifying favorable wettability alterations towards water-wet conditions.

The overall trend observed from zeta potential results supports the premise that Smart Water formulations—especially those incorporating treated produced water—can significantly alter the surface charge in crude oil-brine-rock systems. More negative zeta potentials at brine/rock and crude oil-brine interfaces result in an increasingly electrostatic repulsion between the two interfaces to promote a water-wet system (Jackson et al., 2016; Ayirala et al., 2018). Such favorable wettability alterations are expected to improve the displacement efficiency of oil by lowering the adhesion forces (Li et al., 2020) and increase oil recovery in carbonates.

SmartWater effects on oil recoveries

Coreflooding experiments were performed to evaluate the effectiveness of Conventional SmartWater and two other SmartWater formulations based on Treated Produced Water (TPW1 and TPW2) on oil recovery and pressure drop behavior in carbonate cores. The primary objective was to assess how these different SmartWater chemistries influence residual oil mobilization during tertiary SmartWater injection. In all tests, baseline waterfloods were conducted using high salinity water (regular injection water used in water floods) before the tertiary injection of respective SmartWater solutions, as indicated by the dashed grey line in each plot. The summary of oil recovery data obtained from high salinity water and different SmartWater injections in different core flooding experiments are provided in Table 4.

Table 4—Oil recovery data obtained from waterfloods and different SmartWater formulations used in this study.

Sample	Water Flood (%)	SmartWater Incremental Oil Recovery (%)	Ultimate Oil Recovery (%)
Smart Water 1	59	7	66
SmartWater 2 (TPW 1)	59	16	75
SmartWater 3 (TPW 2)	49	13	62

From conventional SmartWater (SmartWater 1) results shown in Figure 6, oil recovery gradually increased after SmartWater injection, achieving a final recovery of approximately 66% of original oil in place (OOIP). The incremental oil recovery attributed to SmartWater was 7%, over waterflooding. The baseline oil recovery from water flood was 59%. Pressure differential during SmartWater injection peaked around 22 psi, then gradually decreased and stabilized at 15 psi implying residual oil mobilization and propagation in the core. These results suggest that the diluted high-salinity brine (SmartWater 1) was able to mobilize some water flood residual oil and subsequently result in incremental oil recovery.

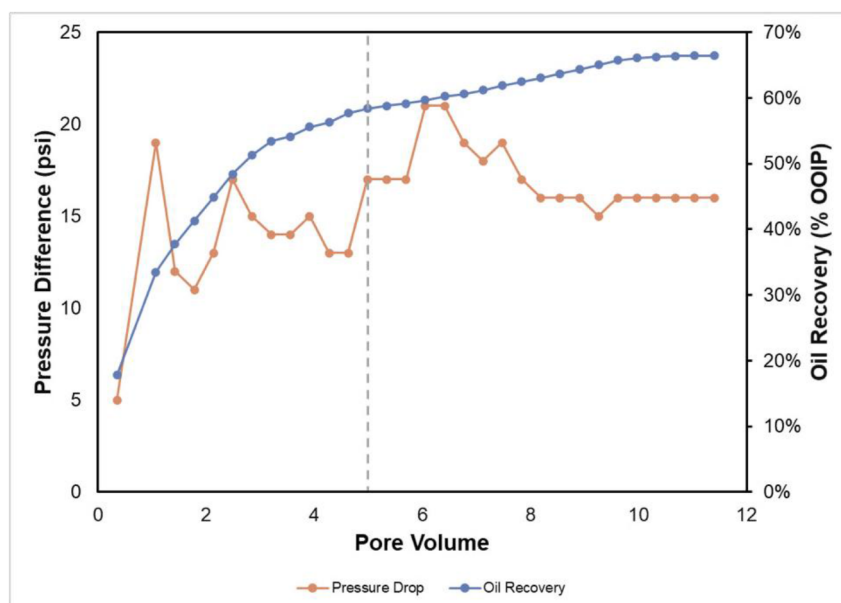


Figure 6—Pressure drop and oil recovery versus pore volume injected for the coreflooding experiment with SmartWater 1. The gray dashed line indicates the injection point of conventional SmartWater.

The oil recovery and pressure drop data obtained from the tertiary injection of SmartWater 2 (TPW 1) in the core are presented in Figure 7. As can be seen, the SmartWater blend significantly enhanced oil recovery. Recovery rose sharply post-injection and plateaued at 75% OOIP, corresponding to an incremental recovery of 16%. The stabilized pressure differential has been found to be relatively lower for SmartWater compared to waterflooding and most of the incremental oil has also been recovered within 1.5 pore volume of SmartWater injection. Such oil recovery behavior can be attributed to stronger rock–fluid interactions and improved wettability shift, which is supported by more negative zeta potential values measured for this SmartWater brine at both carbonate/brine and crude oil/brine interfaces.

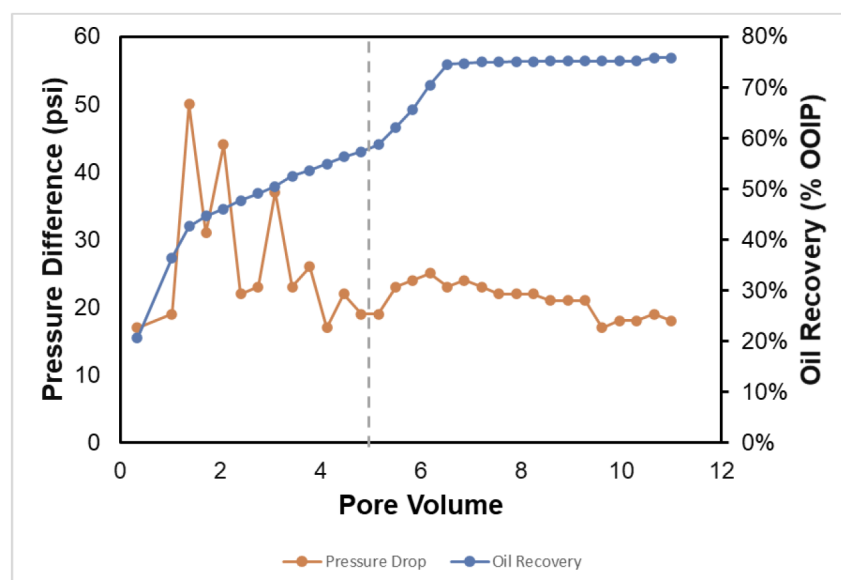


Figure 7—Pressure drop and oil recovery versus pore volume injected for the coreflooding experiment using SmartWater 2 (TPW 1). The gray dashed line represents the point of SmartWater 2 injection.

As shown in Figure 8, SmartWater 2 (TPW 2) also demonstrated effective incremental oil recovery performance, achieving 62% OOIP with an oil gain of 13% over waterflooding. The lower waterflood oil

recoveries obtained from this core flood could be due to stronger native oil-wet conditions. Although the incremental oil recovery was slightly lower than SmartWater 2, SmartWater 3 showed a more stable pressure trend after the initial peak. Such trend indicates more efficient residual oil mobilization for enhanced displacement efficiency in the core, mainly due to the high surface charge alteration capability of the SmartWater 2 formulation at the two interfaces as interpreted from zeta potential measurements

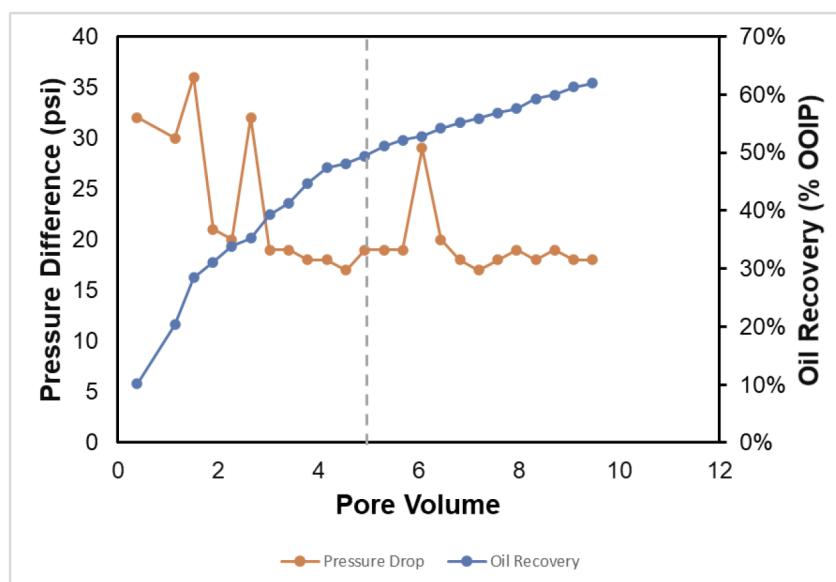


Figure 8—Pressure drop and oil recovery versus pore volume injected for the coreflooding experiment using SmartWater 3 (TPW 2). The gray dashed line represents the point of SmartWater 3 injection.

Overall, the oil recovery results confirm that both SmartWater 2 and SmartWater 3 outperform conventional SmartWater (SmartWater 1) in enhancing oil recovery. The incremental recovery values and pressure behavior trends reinforce the conclusion that ion-tuned treated produced waters can become promising alternatives for SmartWater formulation in carbonate reservoirs. Their ability to promote wettability alteration and improve oil mobilization in carbonates are directly linked with favorable electrokinetic interactions at carbonate/brine and crude oil/brine interfaces.

Summary and Conclusions

This experimental study investigates the influence of SmartWater and treated produced water (TPW) formulations on enhancing oil recovery from carbonate reservoirs through mechanisms of surface charge modification and wettability alteration. The experimental approach involved zeta potential and coreflooding measurements at both ambient (25 °C) and elevated (70 °C) temperatures across three representative interfaces: calcite/brine, carbonate rock/brine, and crude oil/brine.

Zeta potential analysis revealed that SmartWaters based on treated produced water induced significantly more negative surface potentials compared to conventional SmartWater, particularly at calcite/brine and carbonate/brine interfaces. Their surface charge alteration capability with rock further increased at elevated temperature. These results indicate a stronger electrostatic repulsion and more favorable water-wet conditions under elevated temperature, confirming the role of brine composition and ionic strength in modifying mineral surface charges. Although the crude oil/brine interface exhibited less pronounced changes, a similar temperature-dependent trend was observed, indicating enhanced electrokinetic interactions with SmartWaters formulated using treated produced water.

Coreflooding results demonstrated that tertiary injection of SmartWater brines significantly improved the oil displacement efficiency. Conventional SmartWater achieved an incremental oil recovery of 7%

beyond waterflooding, while those based on TPW1 and TPW2 resulted in higher oil gains of 16% and 13%, respectively. SmartWater 2 (TPW 1) achieved the highest incremental recovery factor (16% OOIP), suggesting more substantial wettability alteration and stronger fluid–rock interactions. SmartWater 3 (TPW2), while slightly less effective in increasing recovery over SmartWater 2, demonstrated a more stable pressure response, indicative of effective residual oil mobilization and displacement in the core.

Overall, the findings confirm that treated produced water, when appropriately desalinized and compositionally tailored, can be more effective than conventional SmartWater in modifying surface charge characteristics and improving oil recovery in carbonates. The observed enhancements in wettability alteration and oil displacement efficiency highlights the viability of integrating treated produced water streams with waterflooding to result in sustainable improved oil recovery (IOR) strategies. More importantly, the adoption of such approaches supports closed-loop water management practices in the industry by reducing the reliance on ground water/seawater to promote environmentally responsible use of produced water and circular water economy in oilfield operations.

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